

Controlling water waves with artificial structures

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Abstract

Because of analogies between the 2D Maxwell equations and water wave equations, methods for manipulating electromagnetic waves based on photonic crystals and metamaterials can be extended to manipulate water waves. Doing so provides new opportunities to investigate the interaction of water waves with structures. In this Review, we introduce the research progress of controlling water waves with water wave crystals and metamaterials and summarize the basic theory and calculation methods for water waves. The working principles and design methods for water wave crystals and metamaterials are described, and their properties and applications are presented. We also discuss the current challenges in this field and future directions.

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Key points

- Artificial water wave structures, including water wave crystals and water wave metamaterials, provide an efficient way to control water waves by extending the methods for controlling electromagnetic waves with photonic crystals and electromagnetic metamaterials.
- By engineering the band structures of water wave crystals, interesting phenomenon such as Bloch states, superlensing effect, self-collimating phenomenon and Anderson localization can be observed directly in water waves.
- The band structures of water wave crystals can be calculated by different methods, such as plane-wave expansion method, multiple scattering method, transfer matrix method and finite element method.
- Equivalent material parameters need to be pre-calculated before designing water wave metamaterials with special functionality, and effective medium theory can be used because the water wavelength is much larger than the period of water wave metamaterials.
- Water wave metamaterials have potential applications such as anisotropy of water waves, water wave cloaks, concentration of water waves, isolation and unidirectional propagation of water waves.

Introduction

Water waves are mechanical waves with restoring force provided by gravity and, similar to electromagnetic waves, are classical waves^{1,2}. Water waves are rich in resources as ~70% of the earth is covered by ocean, and ocean wave energy is clean and renewable. Conversely, water waves can destroy equipment floating on the water surface owing to the wave drift force. It is thus important to investigate methods for efficient water wave control, which has potential applications such as water wave energy harvesting³, coastal protection⁴ and offshore structure protection⁵.

The concept of photonic crystals^{6,7}, which control electromagnetic waves, has been extended to acoustics systems^{8,9}. On the basis of band structures corresponding to photonic crystals, many related devices have been designed, such as photonic crystal waveguides^{10,11}, photonic crystal fibres^{12,13} and photonic crystal microcavities¹⁴. In the late twentieth century, researchers paid much attention to metamaterials, which are designed with sub-wavelength building blocks with characteristic dimensions much smaller than working wavelengths. Metamaterials have properties and applications far beyond natural materials, such as negative refraction^{15–17}, zero refractive index¹⁸, spoof surface plasmon polaritons¹⁹, reversed Cherenkov radiation^{20,21}, reversed Doppler effect^{22,23} and superlensing²⁴. With the establishment of the theory of transformation optics^{25,26}, many functional devices based on metamaterials were proposed, including invisibility cloak^{27–29}, rotator^{30,31} and illusion optics³². Except for the aforementioned electromagnetic metamaterials, the concept of metamaterials can also be extended to acoustics^{33–35}, thermal systems^{36,37} and hydrodynamics³⁸.

Because the approximate 2D propagation equation for water surface waves is formally similar to Maxwell's equations for electromagnetic wave propagation, one can extend the success of photonic crystals and metamaterials in regulating electromagnetic waves to

the field of water waves. The interaction of water waves with periodic structures may thereby lead to new phenomena and applications.

In this Review, we review the development of water wave manipulation using periodic structures, that is, water wave crystals and metamaterials. We summarize the fundamental theory of water wave propagation, including the linearization of the water wave equation, the approximation of the water wave equation, the correspondence between electromagnetic and water waves and some classical model structures for controlling water waves. Next, water waves are discussed from the perspective of photonic crystals, analysing the band structures of periodic structures in water waves and the properties associated with the bands. Finally, some basic ideas for the design of water wave metamaterials are presented, and typical applications of water waves are also described.

Basic theory of water waves

Basic equation of water waves

Under the assumptions that the fluid is inviscid, incompressible and irrotational, the propagation of harmonic, linearized water waves is governed by the 3D Laplace's equation³⁹:

$$\nabla^2 \Phi = 0, \quad (1)$$

subject to boundary conditions

$$\frac{\partial}{\partial z} \Phi = \frac{\omega^2}{g} \Phi \text{ on } z = 0, \quad (2)$$

$$\frac{\partial}{\partial \mathbf{n}} \Phi = 0 \text{ on surface } \mathbf{l}. \quad (3)$$

Here, $\Phi(x, y, z)e^{-i\omega t}$ is the velocity potential, ω is the angular frequency, t is time and g is the acceleration due to gravity. The static water surface is on $z = 0$. Surface $\mathbf{l}$ denotes all the interfaces between water and submerged, fixed rigid bodies, and $\mathbf{n}$ is the unit vector normal to surface $\mathbf{l}$.

Equations (1)–(3) are the basic equations to control water waves, valid when the amplitude $\eta = \frac{i}{\omega} \frac{\partial}{\partial z} \Phi(x, y, z=0)e^{-i\omega t}$ of water waves is much smaller than wavelength λ and water depth h . Such equations are a 3D problem and their numerical simulations are usually time-consuming. To overcome this problem, equations (1)–(3) can be approximately reduced to a 2D problem in the x – y plane⁴⁰:

$$\nabla \cdot (u \nabla p) + \frac{\omega^2}{g} p = 0, \quad (4)$$

$$\omega^2 = g u k^2, \quad (5)$$

in which $u = \tanh(kh)/k$, k is the wavenumber, $p = \rho g \eta$ is the hydrostatic pressure at the water surface (in which ρ is the density of liquid) and for convenience, $\nabla = (\partial_x, \partial_y)$ subsequently. Equation (4) is a modification of the commonly used mild-slope equation, suitable for non-uniform systems in which gravity changes with position⁴¹.

Shallow water approximation of water wave equation

When the wavelength of water wave is much larger than the depth of water ($kh \ll 1$), equation (4) can be simplified as¹:

$$\nabla \cdot (h \nabla p) + \frac{\omega^2}{g} p = 0, \quad (6)$$

and the dispersion relation also becomes linear,

$$\omega^2 = g h k^2. \quad (7)$$

Deep water approximation of water wave equation

When the wavelength of water wave is much smaller than the depth of water ($kh \gg 1$), equation (4) can be simplified as:

$$\nabla^2 p + \frac{\omega^4}{g^2} p = 0, \quad (8)$$

and the dispersion relation is:

$$\omega^2 = gk. \quad (9)$$

It shows that the dispersion relation is not affected by the depth in the deep water region.

The relationship between electromagnetic waves and water waves

In the 2D scenario, the transverse magnetic mode of Maxwell's equations in anisotropic materials can be expressed as:

$$\nabla \cdot \left(\frac{\varepsilon}{\det(\varepsilon)\varepsilon_0} \nabla H_z \right) + \mu\mu_0\omega^2 H_z = 0. \quad (10)$$

in which $\nabla = (\partial_x, \partial_y)$, ε_0 is the vacuum dielectric constant, ε is the relative dielectric constant, which can be expressed as a form of tensor

$$\varepsilon = \begin{bmatrix} \varepsilon_x & 0 \\ 0 & \varepsilon_y \end{bmatrix}, \mu_0 \text{ is the permeability of vacuum, } \mu \text{ is the relative permeability and } H_z \text{ is the } z \text{ component of the magnetic field. When one extends equation (4) to anisotropic water depth, the following equation is obtained:}$$

When one extends equation (4) to anisotropic water depth, the following equation is obtained:

$$\nabla \cdot (u \nabla p) + \frac{\omega^2}{g} p = 0. \quad (11)$$

By comparison, it can be observed that equations (10) and (11) are analogous, and the relationship between electromagnetic waves and water waves can be established⁴¹:

$$H_z \leftrightarrow p, \frac{1}{\varepsilon_y \varepsilon_0} \leftrightarrow \frac{u_x}{\rho}, \frac{1}{\varepsilon_x \varepsilon_0} \leftrightarrow \frac{u_y}{\rho}, \frac{1}{\mu\mu_0} \leftrightarrow \rho g, \quad (12)$$

in which h is the static depth (or average depth) of water and can be

expressed as a form of tensor, $u = \tanh(kh)/k$, $h = \begin{bmatrix} h_x & 0 \\ 0 & h_y \end{bmatrix}$. The refrac-

tive index distribution (more exactly, the 'depth tensor') in water waves is analogous to the 'dielectric constant' in optics, and the gravitational acceleration in water waves is analogous to the permeability in optics. The effective water depth^{42–44} and gravitational acceleration^{45,46} can be accurately designed by water wave metamaterials, thus water waves can be fully controlled.

It should be noted that not all the water wave crystals and metamaterials are designed based on the similarity between the shallow water wave equations and 2D Maxwell's equations. In practical applications, such as ocean engineering, nonlinear effects must be considered. The shallow water wave equations provide simple methods to design artificial structures for water wave control, which facilitate the understanding of water wave control in 3D. The schematic diagram of structures, including topographical bottom (Fig. 1a), vertical obstacle structure (Fig. 1b), surface arrays (Fig. 1c), surface plate (Fig. 1d),

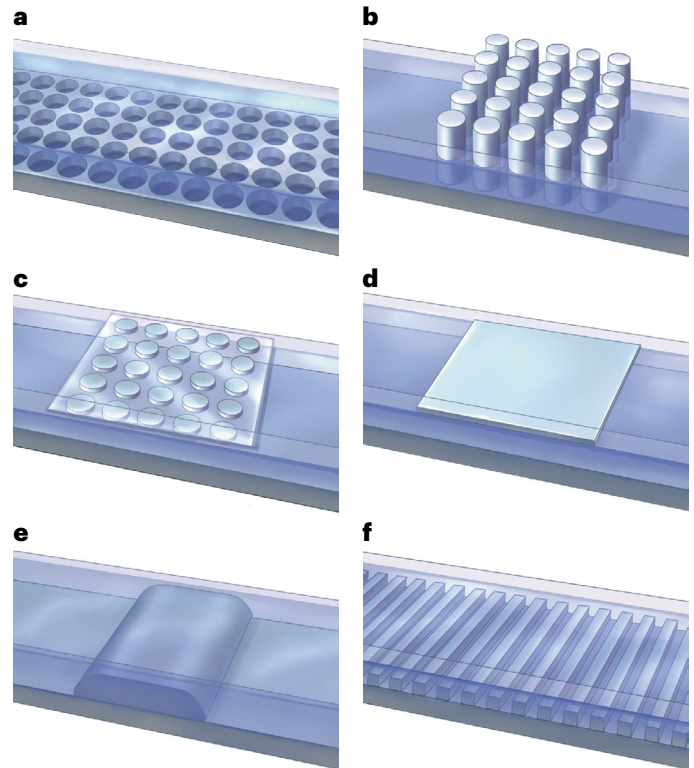


Fig. 1 | Structures for manipulating water waves. a, Topographical bottom. b, Vertical obstacle structure. c, Surface arrays. d, Surface plate. e, Gradient structure. f, Layered structure.

gradient structure (Fig. 1e) and layered structure (Fig. 1f), used to control water waves are shown in Fig. 1. In the following, we mainly focus on introducing water wave crystals and metamaterials based on shallow water equations (SWEs). We also introduce some other structures with the ability to control water waves for comparison.

Evolution of photonic crystals in water waves

Band structure in water waves

Water wave crystals are structures whose periodicity is comparable to or smaller than the working wavelength. Initially, much interest was stimulated by the discovery of the Bragg resonance of water waves when the period of a structure is a multiple of half the water wavelength, resulting in a huge reflection of waves by a finite periodic structure. This phenomenon is not only important for understanding the formation of periodic sandbars near beaches, but also provides a mechanism for coastal protection^{47–51}. Along with the development of photonic crystals, Bloch's theorem has been widely applied to periodic structures in water waves^{52–55}. As the liquid surface waves propagate through periodic structures, they are modulated by the band structures, and bandgaps can also be found in liquid surface waves. Figure 2 shows the band structure and the corresponding transmittance results for a periodic array of cylindrical steps in water waves⁵⁶. Similar to photonic crystals, bandgaps exist between bands of water waves, and even full bandgaps can be found. Simulations and experiments⁵⁷ confirm the existence of bandgaps that inhibit propagation of waves of a specific frequency through such a periodic structure.

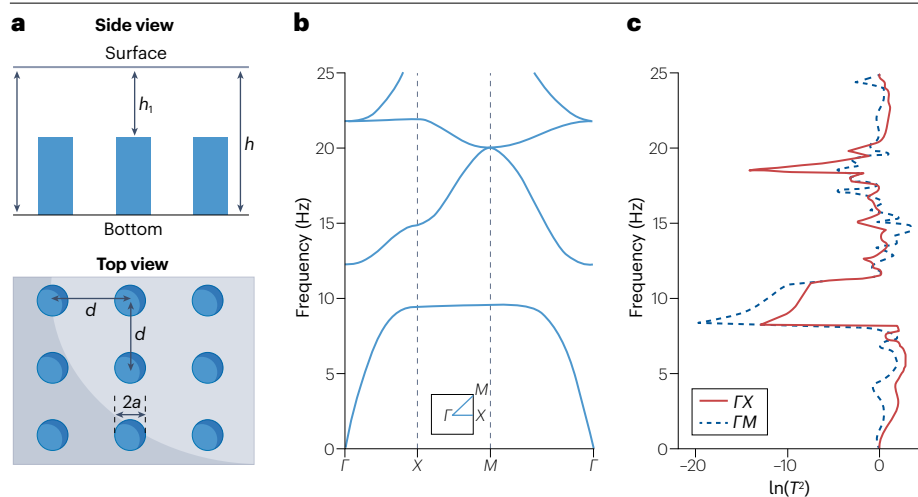


Fig. 2 | A typical periodic structure in water with corresponding band structures and transmittance results. a, An array with lattice constant d of bottom mounted cylinders of radius a and height h in water at a depth of h_1 below the surface. **b**, The band structure corresponds to the periodic structure, the inset shows the Brillouin zone and the direction of wave propagation. **c**, Normalized transmittance $\ln(T^2)$ as a function of frequency for periodic structures in the Γ X and Γ M directions. Figure reprinted with permission from ref. 56, APS.

The location and lattice type of periodic structures in water waves have a strong influence on water waves, and the calculation methods differ for different shapes of periodic structures. Depending on their location in the water, periodic structures can be divided into two main categories: topographical bottoms (including drilled bottoms, bottom mounted cylinders or other vertical barriers)^{55–65} and surface arrays^{66–68}. In terms of dispersion relations, when in the lowest frequency band, topographical bottoms exhibit a slow wave effect in which long water waves are usually slowed down when periodically introduced, whereas surface arrays exhibit a fast wave effect in which the phase velocity and group velocity of water waves can exceed those of the water surface without artificial structures. In addition, when the wavelength is less than the water depth, the surface array is much more sensitive to water wave modulation than the topographical bottom and has a greater effect on water wave propagation. This is also consistent with shallow water surface waves. Depending on the lattice type, periodic structures in water can be classified as square, triangular and graphene-type arrays^{64,69,70}. In the case of periodic cylinders, full bandgaps can be found in all three types of arrays. However, graphene-type arrays can produce more complex bandgaps and can also exhibit bandgaps at low filling ratios, which is more suitable for the practical needs of coastal protection.

Calculation methods for water wave band structure

The main method for calculating the water wave band structure is currently based on the 2D approximation of the water wave equation. For example, the complete bandgap of a liquid surface wave propagating over a periodically drilled bottom is solved by the plane-wave expansion approach⁵⁸, the band structure of a liquid surface wave propagating through a vertical cylindrical array in a square and triangular lattice is calculated by the multiple scattering method⁶⁹, and the omnidirectional total reflection of a liquid surface wave propagating over a bottom with 1D periodic undulations is calculated by the transfer matrix method⁵⁹. It is also possible to solve the exact equations for water waves in 3D directly. For example, the finite element method is used to calculate the full band gap of a stationary surface array of water⁶⁸.

The principles, applicability conditions and characteristics of the methods are detailed in the Supplementary information.

Effects based on water wave band structures

Similar to photonic crystals, there are a number of interesting phenomena based on the band structures of the periodic structures in water waves. In electromagnetic waves, it is difficult to experimentally obtain detailed information on the spatial wave patterns and the evolution of these wave patterns over time; however, the advantage of using liquid surface waves is that these phenomena can be observed visually (Fig. 3). Figure 3a shows the extended Bloch state as the water wave propagates through a square methacrylate box with periodic holes in the bottom. The Bloch state can be visualized when the water wave satisfies a specific frequency⁷¹. In addition, Bloch-like states have also been found in quasi-periodic structures in water waves⁷². Figure 3b shows the superlensing effect of a water wave passing through an array of vertical copper cylinders of a square lattice. The structure has a critical frequency of 5.75 Hz (the first passing band along the XM direction). When the water wave frequency is below the critical frequency, the expected normal refraction is obtained; however, when the frequency is above the critical frequency, negative refraction occurs, leading to the superlensing effect^{73,74}. Figure 3c shows the self-collimation of a point source of water wave passing through a vessel where a flat methacrylate plate with periodically drilled holes is put on the bottom. When the water waves are calm or at low frequencies, the water surface remains stationary or propagates continuously in all directions (Fig. 3c, left two panels). When the water wave frequency lies near the forbidden band and a flat band exists in the constant-frequency surfaces, the shape and length of the ripples along the dominant propagation direction are nearly invariant as they spread, creating a self-collimating phenomenon in water waves⁷⁵ (Fig. 3c, right two panels). Figure 3d shows the localization of water waves over large sand waves with random fluctuations. Laboratory experiments show that Anderson localization can occur through multiple scattering of water waves passing through the bottom of a random pattern^{76,77}. The phenomenon was proved again in a natural sand wave structure by satellite images (Fig. 3d, left half-plane) and photographs of the sea surface (Fig. 3d, right half-plane)⁷⁸. Except for the abovementioned water wave phenomenon, highly directional water wave emission can be realized based on water wave crystals with bandgaps^{79–81}. By placing a water wave point source in the middle of a resonant cavity constructed by two periodically arranged cylindrical arrays with different thicknesses, a highly directional field pattern can

be observed⁷⁹. On the basis of the high density of states at the frequency near the band edge of water wave crystal, highly directional water wave sources can also be realized^{80,81}.

Water wave metamaterials

Methods for designing water wave metamaterials

To design water wave metamaterials with specific functionality, one must pre-calculate the equivalent material parameters. When designing simple structures such as circular structures in water waves, the coherent potential approximation can be applied to find the equivalent parameters as a function of the filling ratio^{45,67,82}. In the case of a cylindrical structure in water, the amplitudes of the water waves inside and outside the effective medium (η' and η'' , respectively) can be derived by applying the column coordinate system with the centre of the cylinder as the origin. Considering the continuity of potential and flow at the effective medium boundary and the zero scattering between the effective medium, the equivalent depth and equivalent gravitational acceleration of the periodic cylindrical structure in the water waves can be derived as a function of the filling ratio. This analytical formulation is of great use in practical designs of metamaterials in water waves.

Unlike the simple structures mentioned earlier, stepped and gradient water wave metamaterials can achieve different equivalent refractive index distributions by changing water depth (Fig. 4a,b), thus

water wave manipulation can be realized at will. When the wavelength is much larger than the period of the structure, effective medium theory (EMT)⁸³ can be used to calculate the effective parameters of structures in water waves. By carefully designing the structures of water wave metamaterials, more complicated functions in water waves can be realized.

Principles and applications of water wave metamaterials

Anisotropy of water waves. Transformation optics plays an important role in the design of optical devices with functionality. To extend the concept of transformation media⁸⁴ in optics to water waves, one needs an anisotropic water depth, which does not exist in nature. In optics, anisotropic materials can be realized by multilayer structures^{84–86}, and the effective parameters can be obtained by the EMT theory⁸³. It has been demonstrated that similar structures can be used to realize anisotropic water depth in water waves, and thus the propagation of water waves can be effectively controlled^{87–90}.

The concept of anisotropic water depth was proposed in the design of a rotator for liquid surface waves⁸⁷, which is a device that can rotate the incoming plane liquid surface wave with a certain angle in the middle of the structure; the idea comes from rotators for electromagnetic waves^{30,31}. The rotator for liquid surface waves was realized by a slanted layered structure arranged in a helical manner (Fig. 5a). Numerical field

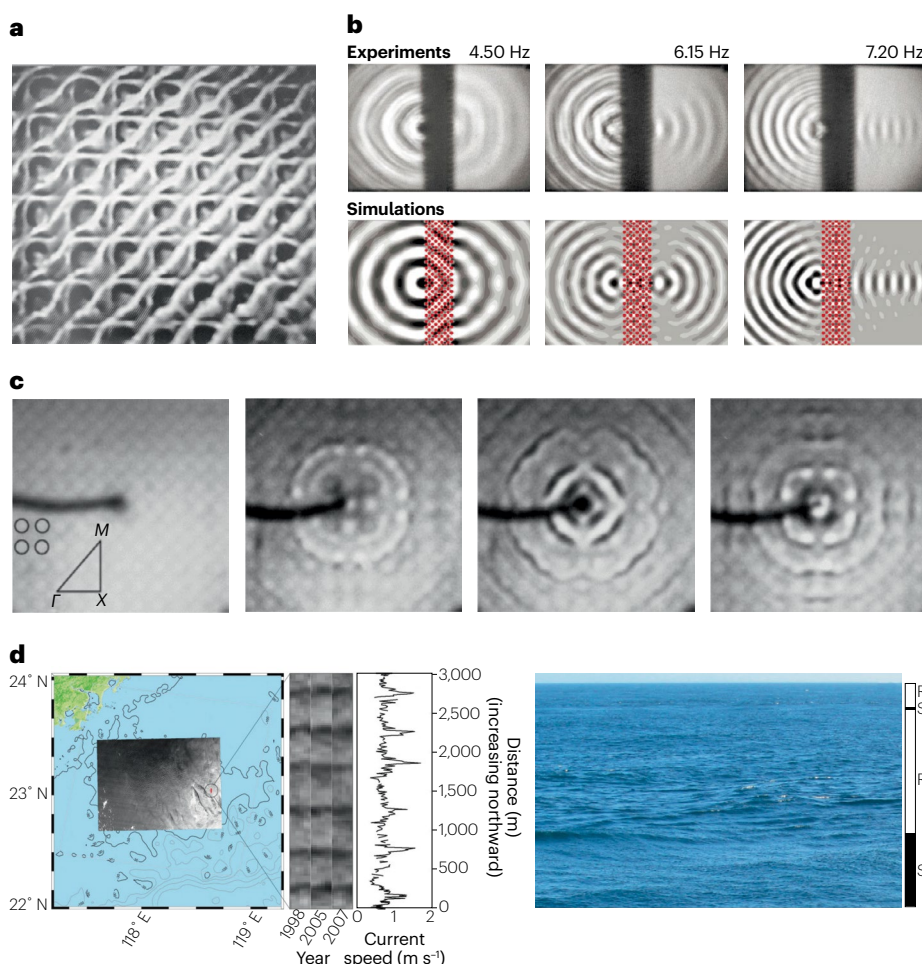


Fig. 3 | Phenomena based on band structures in water waves. **a**, Directly observable extended Bloch state of a water wave passing through a periodic drilled bottom. **b**, Water wave point source refracted through a periodic cylindrical array. Refraction is positive at 4.50 Hz and negative at 6.15 Hz and 7.20 Hz. **c**, Wave pattern of a point source, with frequencies from low to high, passing through the periodically drilled bottom. Self-collimation occurs only near the forbidden band frequencies, as shown in the third and fourth images. **d**, Interaction of water waves with sand waves in the Taiwan Bank. The left half-plane shows a satellite image of the sand waves and the corresponding depth-averaged horizontal velocity, and the right half-plane shows the Anderson localization generated by the structure with the water waves. Part **a** reprinted from ref. 71, Springer Nature Limited. Part **b** reprinted with permission from ref. 73, APS. Part **c** reprinted with permission from ref. 75, APS. Part **d** reprinted with permission from ref. 78, APS.

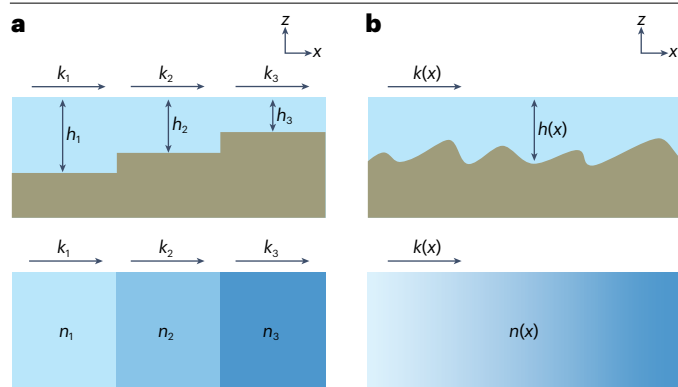


Fig. 4 | Equivalent parameters of metamaterials in water waves. The stepped (part **a**) and gradient (part **b**) water depth distributions realize the stepped and gradient refractive index distribution of water waves.

patterns (Fig. 5b) show that when a plane wave passes through the rotator at $f = 6$ Hz from the left side, the wave in the middle rotates a certain angle, with no scattering. The corresponding experimental result (Fig. 5c) also shows that waves in the middle region rotate accordingly, but the wave form is weakened, resulting from the dissipation caused by the interaction between fluid and solid⁸⁷.

Water wave metamaterials with anisotropic water depth can also be used to design reflectionless bending waveguide for liquid surface wave⁸⁸. A designed waveguide with bending angle $\theta = 20^\circ$ is shown in Fig. 5d, and the tilted angle of the multilayer structures with an effective anisotropic water depth is $\alpha = 32.84^\circ$. When water wave incidents from the left port at $f = 4$ Hz, experiments show that for an empty bending waveguide, the scattering is large, whereas for a bending waveguide with metamaterials, the scattering can be eliminated (Fig. 5e). By using a Fourier transform, the contribution of the zero-order waveguide mode to the total field in the transmission area ($x > 0$) and reflection area ($x < 0$) separated by the bending boundary ($x = 0$) can be obtained (Fig. 5f). It can be seen that the waveguide with metamaterials can well suppress high-order modes and maintain a high proportion of zero modes. Therefore, the scattering of the transmission field is greatly reduced, such that the plane wave can maintain a relatively complete waveform after turning⁸⁸.

It is reported that the actual anisotropy of a layered bathymetry is larger than that obtained from 2D SWE; this is because the 3D near field effect cannot be effectively demonstrated by 2D SWE⁸⁹. The experimental diagram is shown in Fig. 5g, with bottom metamaterial structure consisting of 60 layers of glass and a point source generated by an oscillator. The experimentally measured field patterns show the anisotropy of the structure (Fig. 5h). By comparing water wavelength (including λ_x and λ_y) obtained by a full 3D homogenization theory⁹⁰ and the experimental results, it can be concluded that to obtain effective water depth, 3D near field effects need to be considered⁸⁹ (Fig. 5i). In addition, with alternatively arranged open canals and resonant canals in the design, a hyperbolic water wave metamaterial with an effective negative water depth along one direction was theoretically proposed, and negative refraction with a hyperbolic dispersion was also numerically observed⁹¹.

Water wave cloaks. Optical invisibility cloaks have been developed based on transformation optics^{92,93}, and the method of

designing invisibility cloaks has been extended to other field dynamics⁹⁴. In addition, water wave cloaks have considerable application prospects, such as protecting buildings from water waves, hiding military installations and so on. Thus, it is worth exploring the design methods for water wave cloaks^{95,96}.

A cloak for linear surface waves was realized by using rigid fan-shaped structures with an effective inhomogeneous anisotropic viscosity in a cylindrical coordinate system⁹⁷. The cloak consists of 100 rigid fan-shaped sectors periodically arranged around the centre, and its inner radius is 41 mm and its outer radius is 100 mm (Fig. 6a). When liquid surface waves pass through the cloak, a cloak regime is formed in the centre. When a point source is placed outside the cloak, reflection is greatly attenuated at the frequency of 10 Hz (ref. 97) (Fig. 6b).

To reduce the difficulty of experiments on water wave cloaks, water wave carpet cloaks based on transformation optics were proposed^{98–100}, in which the required parameters are simplified. A designed carpet cloak is made of 10 layers PLC plastic with alternate depth (Fig. 6c), which is equivalent to an anisotropic water depth. The cloak area of the designed carpet cloak is a triangle area with a slant angle of α . Numerical results show that when a Gaussian beam impinges on the carpet cloak with an incident angle of 30° at a frequency of 10 Hz, the field pattern of the reflected wave is the same as it impinges on a flat plane (Fig. 6d). The experimental results (Fig. 6e) agree with numerical results, which show a favourable performance of the designed carpet cloak⁹⁸. In addition, by controlling the local reflection phase of the unit cell, a metasurface carpet cloak for water waves can also be realized by compensating the phase distortion⁹⁹.

The bandwidth of the aforementioned water wave cloaks designed based on transformation optics is narrow, which may not suffice for practical applications. In optics, when graded refractive index metamaterials are introduced into waveguide systems, broadband waveguide cloaks based on mode evolution can be realized¹⁰¹. By extending the mechanism of mode evolution to water waves, a broadband water waveguide cloak was realized¹⁰². The designed water waveguide cloak consists of a planar waveguide attached by graded refractive index materials (Fig. 6f, top), and the experimental realization is shown in the bottom of Fig. 6f. The numerical results of the water waveguide cloak with the water depths of 0.15 m and 0.16 m at the working frequency of $f = 0.7$ Hz are shown in the upper part of Fig. 6g. The cloaking performance is good. The physical mechanism is that when a water wave is incident from the left port, the propagating wave is gradually converted into surface modes and then released as propagating waves, which creates a cloaking area in the middle of the waveguide cloak. When an object is placed in this area, owing to the evanescent propagation properties of the surface modes, the object does not affect the propagation of waves in the waveguide. Experimental results for the case of $h = 0.16$ m at $f = 0.7$ Hz is shown in Fig. 6g (bottom) and are consistent with the numerical results¹⁰². In a similar way, a floating object can be cloaked by careful design of the bottom topography, by which the incident water wave can enter into the underneath of the floating object and then recovered on the water surface¹⁰³.

Cloaking can also be realized by the scattering cancellation method, in which the scattering of structures in the far field can be cancelled out. For shallow water waves, a vertically pierced cylinder can be cloaked by variable bathymetry¹⁰⁴. However, when it comes to deep water, variable bathymetry would not work. It was theoretically demonstrated that a vertically truncated cylinder in deep water can be

cloaked by careful design of a circularly arranged truncated vertical cylinder or toroidal ribs^{105,106}, as later verified by experiments¹⁰⁷. By using a real-coded genetic algorithm^{108,109}, the cloaking configurations for a certain frequency and direction were obtained for four and eight circularly arranged cylinders, and the effect of the wave direction was studied numerically and experimentally¹⁰⁸. It was also demonstrated that the cloaking configurations can be used to reduce the wave drift force for the structures¹¹⁰. In addition, a broadband cloak with reduction of wave drift force can be realized by the arrangement of location defects^{111,112}.

Artificial intelligence¹¹³ plays an important role in the design of water wave devices, and it breaks the limitation of traditional water wave cloaks, such as narrow bandwidth and extremely complex geometrical parameters. By using the topology-optimized method, a new kind of water wave cloak was designed, which was realized by inversely designing the water depth distribution outside the cloaking area¹¹⁴. By optimizing the irregular structures between the inner and outer radii of the cloak by the topology-optimized method, the scattering of the obstacle can be cancelled. The cylindrical cloak has an inner radius of 40 cm and an outer radius of 116 cm. Numerical and experimental

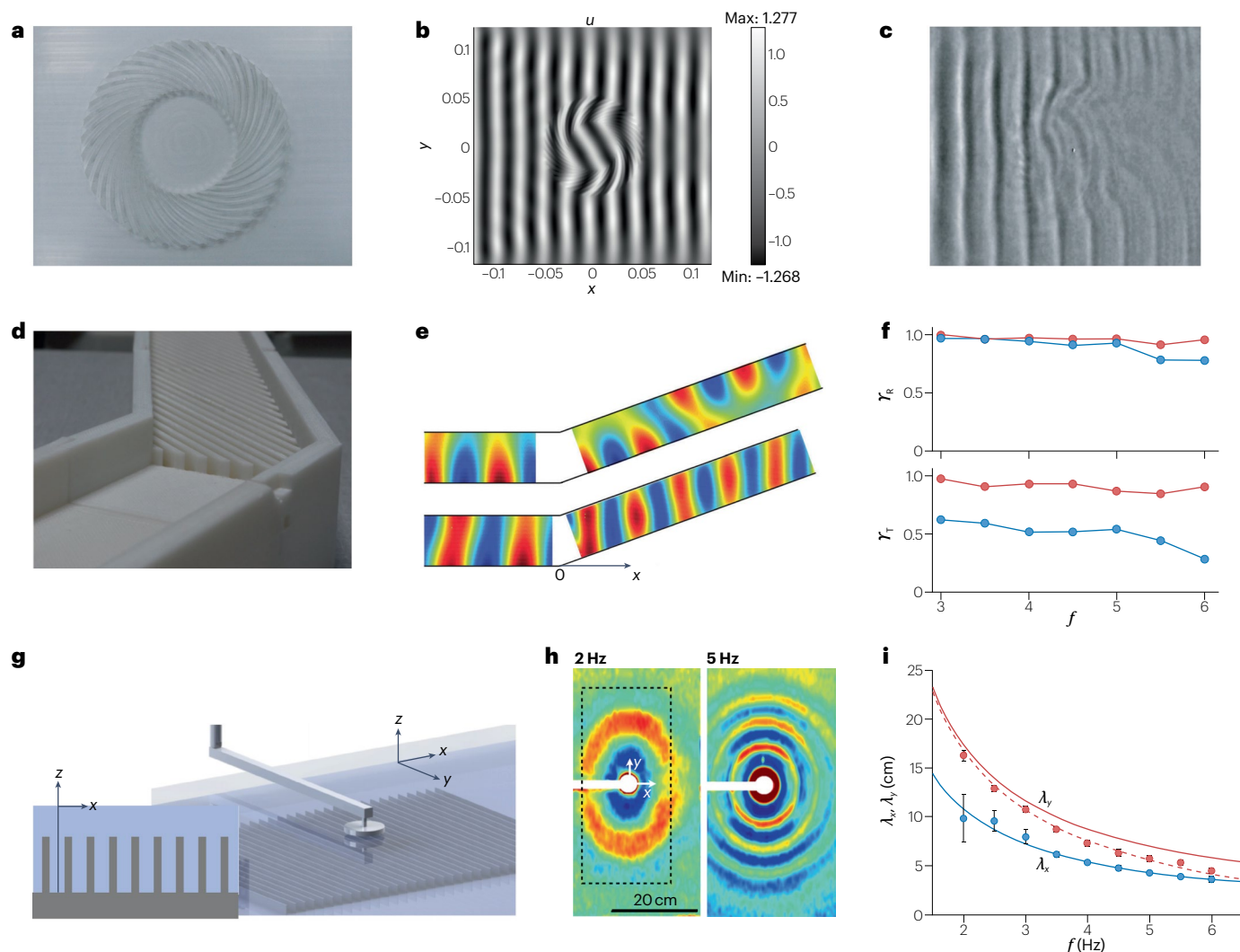


Fig. 5 | Water wave metamaterials with anisotropic water depth. **a–c**, Rotator for liquid surface waves. A liquid surface wave rotator consisting of 72 helical oblique layers (part **a**). The numerical (part **b**) and experimental (part **c**) results for a rotator at 6 Hz. **d–f**, Bending waveguide for liquid surface waves. Bending waveguide with metamaterials (part **d**). Experimental results (part **e**) for an empty waveguide (upper) and a waveguide with metamaterials (lower) at 4 Hz, in which a wave is incident from the left port. Contribution of the zero mode to the total field with respect to the frequency f (part **f**), in which the black line for the empty waveguide, and the red line for the waveguide with metamaterials. Here, r_r and r_t represent the contribution of the zero mode to the total field in the transmission area ($x > 0$) and reflection area ($x < 0$), respectively. **g–i**, Realization

of water wave anisotropy. Anisotropic metamaterials made of 60 layers of glass and a point source generator (part **g**). Experimental results for a point source placed in the middle of the structure at frequency of 2 Hz (left) and 5 Hz (right) (part **h**). Wavelengths of water waves as a function of f , where symbols represent experimental results, plain lines represent theoretical results based on 3D homogenization theory in shallow water regime and dashed lines represent the theoretical results obtained by 3D homogenization theory with dispersive correction (part **i**). λ_x and λ_y are wavelengths of water waves along x and y directions, respectively. Parts **a–c** reprinted with permission from ref. 87, IOP. Parts **d–f** reprinted with permission from ref. 88, APS. Parts **g–i** reprinted with permission from ref. 89, APS.

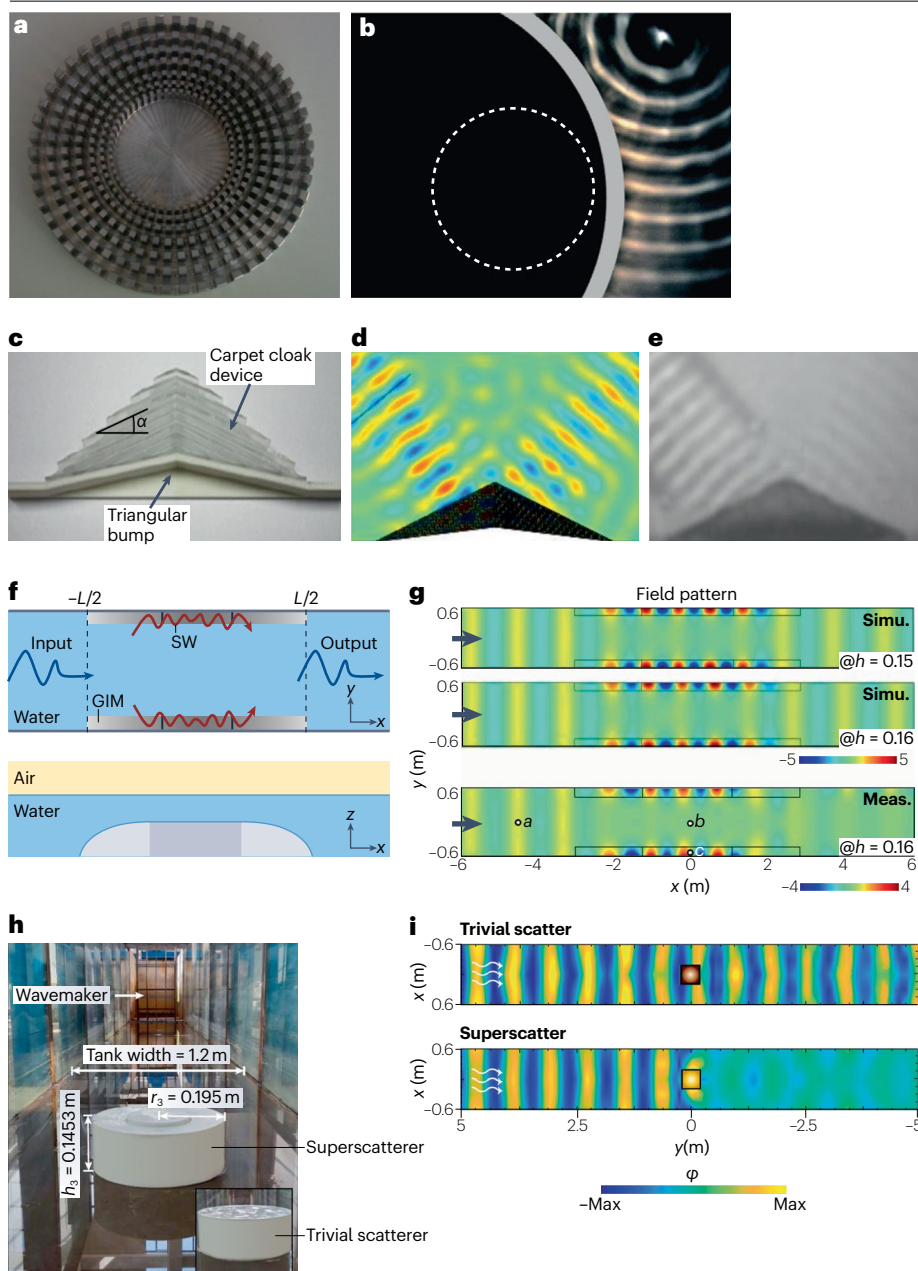


Fig. 6 | Water wave cloaks and superscatters. **a,b**, An annular cloak consisting of 100 rigid fan-shaped unit cells, with inner radius 41 mm and outer radius 100 mm (part **a**) and a snap shot when the frequency of a point source is 10 Hz. **c–e**, A carpet cloak consisting of 10 layers with alternative water depth (part **c**) along with numerical (part **d**) and experimental (part **e**) results at 10 Hz with an incident angle of 30°. **f,g**, Water waveguide cloaks. The top view and side view of a waveguide cloak consist of graded refractive index materials (part **f**). The simulated (simu.) and measured (meas.) field patterns of waveguide cloaks at 0.7 Hz with different water depths h . **h,i**, Superscattering of water waves. The experimental setup for the water wave superscatter (part **h**). The experimental field patterns (part **i**) of the superscatter (left) and the trivial scatter (right). Parts **a** and **b** reprinted with permission from ref. 97, APS. Parts **c–e** reprinted with permission from ref. 98, APS. Parts **f** and **g** reprinted with permission from ref. 102, APS. Parts **h** and **i** are reprinted from ref. 117, CC BY 4.0.

results both show that when a plane wave passes through the cloak, the field pattern of the incident wave is not affected by the obstacle¹¹⁴.

Unlike cloaking, which aims to eliminate the scattering crossing section, superscattering^{115,116} aims to enlarge the scattering crossing section. By using multilayered concentric cylinders with an inhomogeneous depth profile, the long-held single-channel scattering limit can be broken by several times and the total scattering strength of water waves can be enhanced, and superscattering of water waves can be realized¹¹⁷. The corresponding experimental setup is shown in Fig. 6h (middle). Different heights are used to achieve different water depth profiles. The cylindrical structure (with no special structure on the surface) used for comparison is shown in Fig. 6h (lower right corner),

and the experimental results are shown in Fig. 6i. The left part is the experimental group for superscatter, and the right part is the reference group. Water waves with frequency $f = 1.55$ Hz are incident from above, and the results indicate that the superscatterer isolates the wave and greatly reduces the amplitude of the water wave behind it¹¹⁷.

Concentration of water waves. In ocean engineering, the concentration of ocean waves is important for harvesting³ and conversion^{118,119} of ocean wave energy. Thus, a reasonable arrangement of water wave energy concentrating devices to collect ocean wave energy is crucial. In addition, it is important to design energy concentration devices with high energy collection efficiency.

Wave focusing plays an important role in the concentration of water waves and can be used to realize water wave energy conversion^{118,119}. Traditionally, the focusing of water waves is realized by water wave lenses, by using structures with a shape of Fresnel type¹²⁰, biconvex^{121,122}, ellipse¹²³, parabolic^{124,125}, crescent¹²⁶ and so on. To further improve the wave energy conversion efficiency, a water wave lens can be integrated with wave energy converters¹²⁷. However, traditional water wave lens designed based on the principles of refraction or reflection shows limitation for the concentration of water waves and new methods are desired. Metamaterials provide a new way to realize water wave lenses. In the following, we introduce some water wave lenses based on metamaterials.

In liquid surface waves, with the concept of metamaterials, it is demonstrated that the effective depth and effective gravity depend on the filling factors of the bottom-mounted cylinders in the long wavelength limit. On the basis of this property, a focusing lens for liquid surface waves was theoretically proposed by designing a biconvex periodic array, which consists of cylinders⁸². Later, an experimental demonstration for the focusing effect of the biconvex periodic array for liquid surface waves was conducted. When a plane liquid surface wave with $f = 7.77$ Hz impinges on the designed lens, the focusing effect of the experimental result is evident¹²⁸ (Fig. 7a). By using a gradual filling fraction of cylinders, a gradient refractive index lens with flat surface

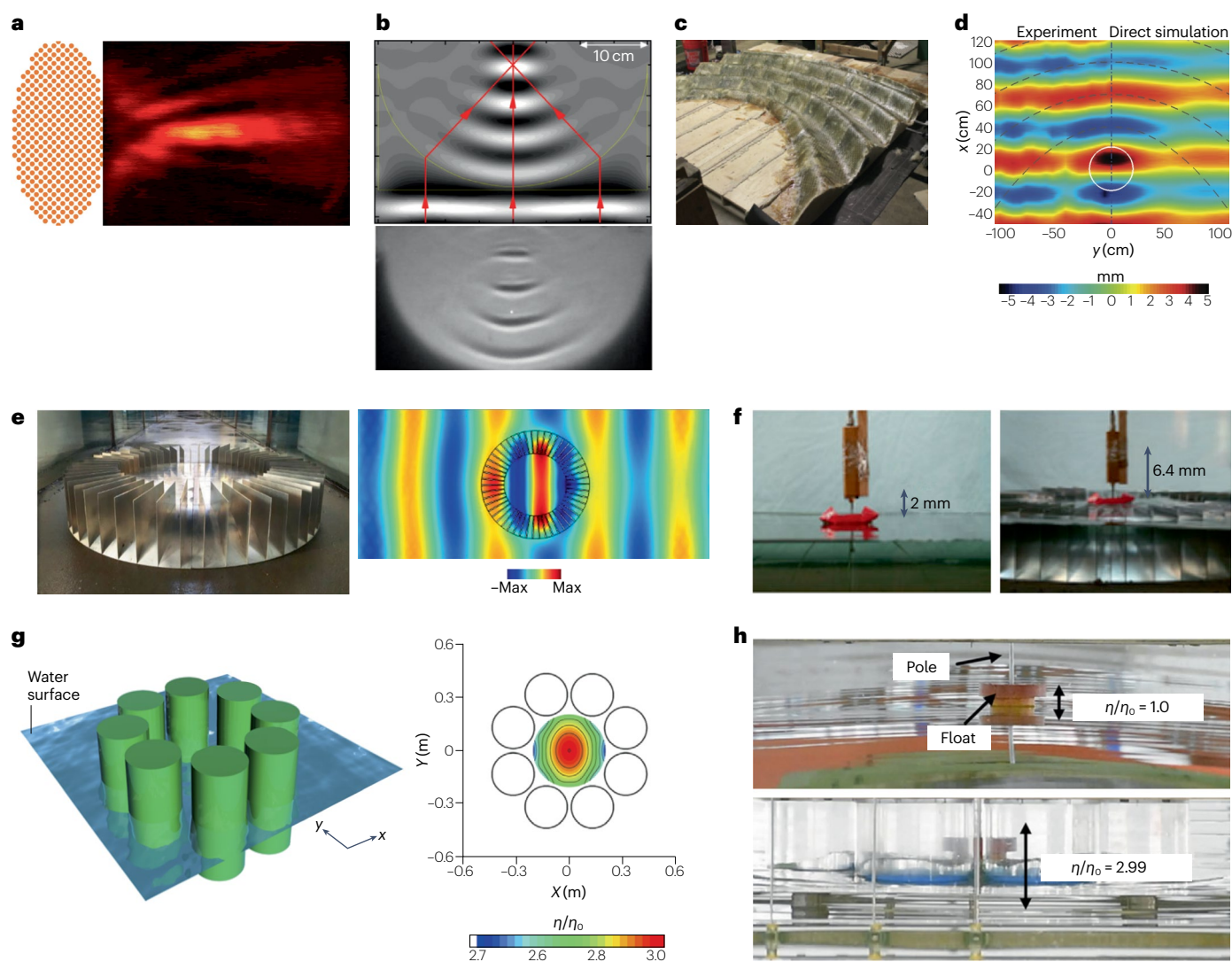


Fig. 7 | Water wave lens and concentrators. **a**, Water wave lens with periodically arranged structures. Experimental results (right) of a surface water wave lens (left) at $f = 7.7$ Hz. **b**, Numerical (top) and experimental (bottom) results of a zero-index lens at 4 Hz. **c,d**, Water wave lens based on Bragg resonance. A surface water wave lens composed of smooth curved seabed ripples (part **c**). The numerical and experimental results of a surface water wave lens at $\lambda = 60$ cm (part **d**). **e,f**, Concentrators based on Fabry–Pérot resonances. A concentrator made of dam-boards at $f = 1.5$ Hz (part **e**) and the corresponding simulation results. The experimental results (part **f**) of the case with (right) and without (left) a concentrator

for a plane wave at $f = 1.5$ Hz. **g,h**, Cylindrical energy concentrator. Schematic of the designed cylindrical energy concentrator (left) and simulation results (right) for a cylindrical energy concentrator composed of eight circular arranged discs at 0.91 Hz (part **g**). Experimental results (part **h**) of the case with (bottom) and without (top) the cylindrical energy concentrator¹³⁹. η , real-time amplitude of water waves; η_0 , amplitude of incident water waves. Part **a** reprinted with permission from ref. 128, AIP. Part **b** is reprinted from ref. 133, CC BY 4.0. Parts **c** and **d** reprinted with permission from ref. 136, APS. Parts **e** and **f** reprinted with permission from ref. 138, APS. Parts **g** and **h** reprinted with permission from ref. 139, AIP.

for linear surface waves was experimentally realized¹²⁹, which is different from the lens whose focusing effect is based on a curved surface^{82,128}. Another way to realize effective gradient refractive index in water waves is based on the gradient isotropic water depth^{130,131}, and a water wave lens with a refractive index profile of Maxwell fisheye lens¹³² has been experimentally demonstrated¹³⁰.

A special kind of metamaterials in optics is zero index metamaterials¹⁸, and the concept can be extended to water waves as well. A rigid plate placed on the water surface can be effectively treated as a zero-index metamaterial¹³³. When the designed rigid plate on the water surface has a concave shape, water waves can be focused. Both numerical and experimental results show the focusing effect for an incident water wave with $f = 4$ Hz passing through the rigid plate with an effectively zero index¹³³ (Fig. 7b). Similarly, epsilon-near-zero metamaterials¹³⁴ can also be extended into water waves, and an epsilon-near-zero lens with a semicircle shape for water wave has been experimentally realized¹³⁵. In addition, the Bragg resonance caused by bottom modulations of the seabed can be used to focus water waves¹³⁶. Figure 7c shows the wave tank with smooth curved seabed ripples in experiments. Both numerical and experimental results show the incident water wave can be focused at the designated position (Fig. 7d) at the wavelength of 60 cm. By designing polychromatic seabed topography, a broadband focusing effect can be realized¹³⁶.

Water wave concentrators can also be used to realize the concentration of water wave energy, using a different physical principle to the water wave lens. By introducing the mechanism of Fabry–Pérot resonance¹³⁷ into the transformation design of water wave devices, a water wave concentrator can be realized¹³⁸. The water wave concentrator can eliminate the scattering of waves and can collect the wave energy at multiple resonant frequencies. Figure 7e (left) shows the structure of the water wave concentrator, which consists of circularly arranged dam-boards with gradient water depth along radial directions. As waves are confined in the dam-boards along the azimuthal direction, the concentrator possesses an extremely anisotropic water depth tensor. Numerical simulations show that when a water wave with $f = 1.5$ Hz passes through the concentrator, the amplitude of the water wave in the middle of the concentrator is amplified (Fig. 7e, right). Experimental results (Fig. 7f, left) show that when a small boat model is placed on the water surface without the concentrator, the oscillating amplitude of the boat is 2 mm, which is the same as the amplitude of the incident wave. Although the boat model is placed in the middle of the water wave concentrator (Fig. 7f, right), the oscillating amplitude of the boat model is amplified to 6.4 mm, which demonstrates the functionality of the concentrator¹³⁸.

A different mechanism underlies a water wave energy concentrator based on cylindrical discs arranged in circular columns. When the scattering between each cylinder can eliminate the scattering of the whole cylindrical energy concentrator, all the energy enters into the concentrator, which leads to the amplification of waves^{139,140}. The cylindrical energy concentrator is composed of eight circularly arranged cylindrical discs (Fig. 7g, left). The numerical result shows that the maximum water wave amplitude inside the concentrator at a frequency of $f = 0.91$ Hz is 2.91 times larger than the amplitude of an incident water wave (Fig. 7g, right). In experiments using an incident wave with a frequency of $f = 0.91$ Hz, when there is no cylindrical energy concentrator, the amplitude of the incident wave is not changed (Fig. 7h, top), and when a cylindrical energy concentrator is placed in the water (Fig. 7h, bottom), the amplitude of water wave at the centre of the concentrator is amplified 2.99 times¹³⁹. With the resonant effect of the

structures, the concentration of water waves can also be realized. By the design of graded bottom-mounted resonant arrays, following the concept of rainbow trapping^{141,142} in optics, water waves can be localized and amplified at specific locations in a wide frequency range^{143,144}.

Isolation and unidirectional propagation of water waves. The propagation of water waves in the ocean is disordered, so an efficient way to control the propagation of water waves is needed. Metamaterials can be used to control the propagation of water waves and even isolate water waves, which has an important role in the protection of marine equipment.

It was theoretically predicted that by introducing resonant splitting tubes into the design of water wave metamaterials, effective negative gravitational acceleration can be realized, and water wave isolation can be achieved⁴⁵, without relying on the bandgap of water wave crystals at higher frequencies. Later, this theoretical prediction was experimentally verified⁴⁶. The measured transmission spectrum is consistent with the results from EMT and simulations (Fig. 8a), and the isolation of water waves is demonstrated by the existence of a transmission dip caused by the effective negative gravity⁴⁶. By introducing the concept of the Autler–Townes splitting resonance¹⁴⁵ to water waves, perfect absorption without reflection and transmission can be realized in a water wave guide with two resonant side channels, which can be used to isolate water waves¹⁴⁶. The isolation of water waves by the ancient Luoyang Bridge has also been interpreted in terms of metagratings¹⁴⁷. Owing to the similarity between the structure of realistic bridges and optical metagratings, the underlying physical mechanism behind the isolation effect of realistic bridges can be investigated by introducing the theory of optical metagratings to water waves, which will further contribute to the design of bridges.

By breaking the spatial inversion symmetry of structures, unidirectional transmission of water waves can be realized, which also yields the ability to isolate water waves^{67,148,149}. It is reported that 1D periodic undulations together with a segmented breakwater can be used to construct a water wave diode, possessing unidirectional transmission of water waves. Numerical results (Fig. 8b) show that the water wave diode can only block waves incident from the right side¹⁴⁸. In addition, with the design of rigid periodic disc arrays covering on the water surface, a fast water wave system was proposed. Because the effective refractive index of the designed structure is in the range of $0 < n < 1$, for water waves with incident angle above the critical angle, total reflection can be achieved. By arranging the array discs on the water surface in an isosceles right triangle, unidirectional transmission of water waves can be realized. Numerical and experimental results show that the incoming water wave from the left side is isolated (Fig. 8c), and that from the right side is transmitted¹⁴⁷ (Fig. 8d). Another way to isolate water waves is to introduce loss into a Willis metamaterial. The unit cell of the Willis metamaterial consists of a crescent pedal and Willis coupling can occur when the mirror symmetry of the unit cell is broken (Fig. 8e). By introducing loss, asymmetric reflection for the forward and backward incident water waves can be achieved¹⁵⁰.

Owing to applications in the transportation of marine cargo and collection of marine refuse, unidirectional propagation of water waves is important in water wave control. For a 1D groove array (Fig. 8f, left) vertically placed in water, its effective water depth is negative, and it supports surface water wave polaritons⁴¹. Numerical results show that when the structure is excited by a vortex source with symmetry breaking, unidirectional propagation of water wave polaritons can occur (Fig. 8f, right). Experimentally (Fig. 8g), it can be seen that

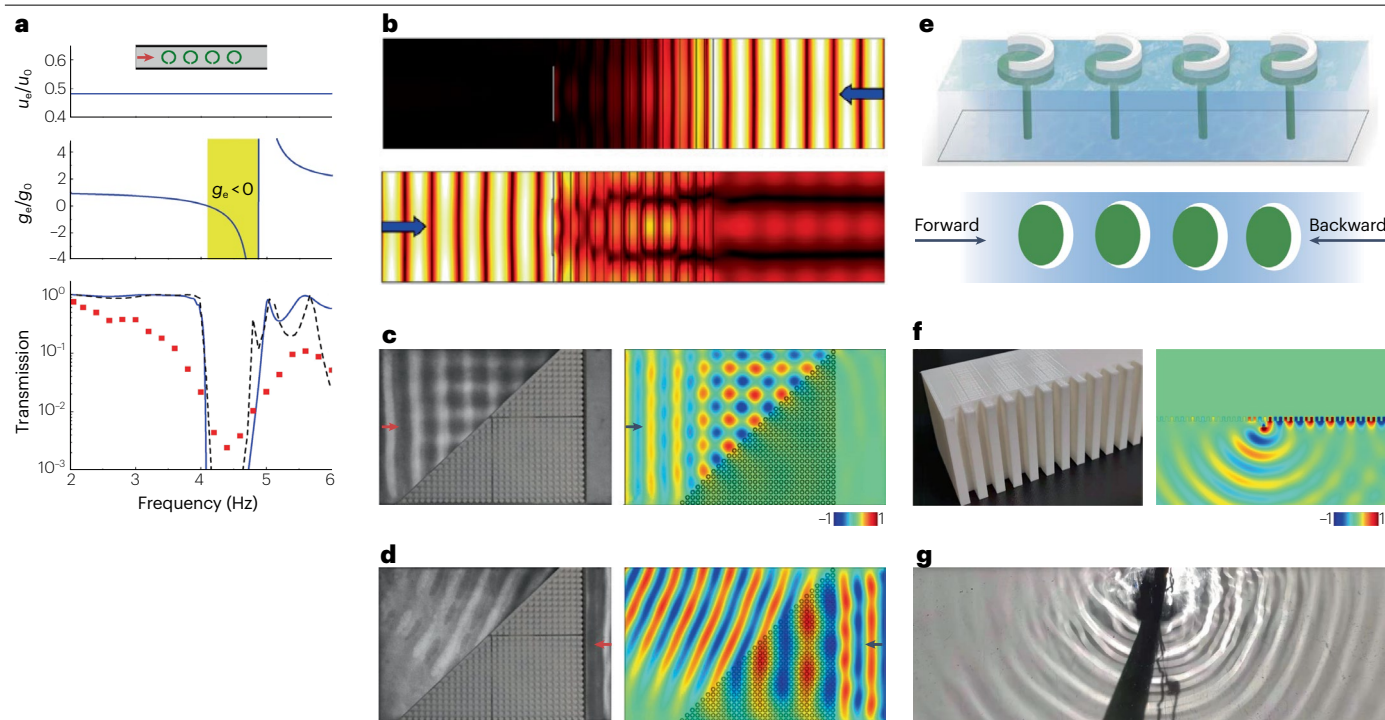


Fig. 8 | Isolation and unidirectional propagation of water waves. **a**, Effective depth (u_e/u_0), effective gravitational acceleration (g_e/g_0) and transmission results corresponding to the experiments (red squares), theory (solid lines) and simulations (dashed lines) for an array of splitting tubes (inset). **b**, Unidirectional propagation of water wave diode. **c,d**, Experimental and numerical field patterns for periodic disc arrays covering on the water surface with a shape of isosceles right triangle with the water wave incident from left (part c) and right (part d). **e**, Asymmetric propagation of lossy Willis water wave metamaterials.

f,g, Unidirectional propagation of water waves. One-dimensional groove array (left) with effective negative water depth and the numerical field pattern (right) of unidirectional water wave polaritons (part f). The experimental field pattern of unidirectional water wave polariton with a vortex source with a wavelength of 5 cm placed near the groove array (part g). Part a reprinted from ref. 46, Springer Nature Limited. Part b reprinted with permission from ref. 148, IOP. Parts c and d reprinted with permission from ref. 67, APS. Part e is reprinted from ref. 150, CC BY 4.0. Parts f and g reprinted with permission from ref. 41, APS.

when a clockwise vortex wave with a wavelength of 5 cm placed near the structure, water wave polaritons only propagate along the right direction⁴¹. It was reported that unidirectional propagation of water waves can be experimentally excited near the Luoyang Bridge by vortex sources¹⁵¹. In addition, this phenomenon was theoretically demonstrated by analytically derived surface mode of water waves based on EMT. Compared with the reflection-type 1D groove array used to verify the unidirectional water wave polaritons, the Luoyang Bridge can be regarded as a transmission-type water wave metagrating with large size, and the nonlinear dispersion of water waves should be considered¹⁵¹.

Outlook

Although complete theoretical and experimental systems have been developed for the control of water waves by wave crystals and metamaterials, there is still one problem that needs to be solved: with the increase of water depth and water wave amplitude, nonlinear effects^{120,152,153} have a substantial influence on the performance of water wave devices. Such effects are key to water wave control devices from laboratories to practical applications. Only a few works have paid attention to this point^{154,155}, thus nonlinear effects on water wave control are an important future direction in the field of water wave control.

One kind of water wave metamaterials, time-modulated water wave metamaterials, has not been introduced in the main text as there is relatively little associated research. At present, only time

inversion focusing has been realized for water wave time-modulated metamaterials^{156,157}, but in other wave domains, time-modulated metamaterials have been used for functions such as negative refraction¹⁵⁸, imaging¹⁵⁹, non-reciprocal propagation¹⁶⁰, anomalous reflection and transmission¹⁶¹ and so on. Therefore, this is also an important direction of water wave control.

As the 2D counterpart of metamaterials, metasurfaces^{162–165} have attracted attention because their structures are simple and they are readily fabricated. As a special kind of metasurfaces, metagratings^{166–168} with thickness near one wavelength also possess abilities in wavefront control with a freedom of phase gradient. When the concept of metagratings is introduced into water waves, the wavefront of water waves can also be controlled¹⁶⁹. We hope more water wave devices based on metagratings would be realized in the near future.

With the development of topological photonics¹⁷⁰, the concept of topology has been widely used in classical wave systems, which promotes the understanding of wave propagation. Topological states can also be realized in water wave systems. By introducing the concept of topology to water waves, the existence of a topological transition of 1D water wave systems for shallow, intermediate and deep water waves was theoretically confirmed by changing the width of the channel¹⁷¹. The topological interface state at the boundary of two periodic arrays with different configurations was also theoretically demonstrated¹⁷¹. Furthermore, the topological interface state was experimentally

observed in a water channel at the interface of a heterojunction constructed by two periodic sidewalls with different Zak phases¹⁷². For 2D water wave systems, it was theoretically reported that by designing the water bottom with height periodically modulated in a triangular pattern, a topological helical edge state with immunity to defects and disorders can be realized¹⁷³. With a hexagonal water wave crystal, topological edge states at the domain walls in intermediate gravity–capillary waves with dispersion were experimentally demonstrated¹⁷⁴. Later, these topological edge states were also experimentally observed in a square lattice system¹⁷⁵. Note that the research about topological phenomena in water waves has developed slowly in recent years. We believe that it is worth exploring topological phenomena in water waves in the near future, which will facilitate the development of water wave applications, such as water wave energy harvester.

Bound states in the continuum¹⁷⁶ are localized states with large field enhancement and infinite quality factor lying in the continuum regime of the spectrum, which were first proposed in quantum mechanics¹⁷⁷ and later extended to optics^{178,179}, acoustics¹⁸⁰ and elastics¹⁸¹. Trapped modes^{182–186} in water waves now can be revisited in the light of bound states in the continuum.

Spin is a basic property of quantum particles¹⁸⁷, but it has been demonstrated that spin angular momentum exists in electromagnetic fields^{188,189}, acoustic fields^{190,191} and elastic fields¹⁹². The spin of water waves has been proposed and experimentally demonstrated¹⁹³. In addition, by using the interference of two standing water waves, the vorticity of water waves can be observed^{194,195}. Control over water wave spin by tuning wave source shows limitations, but introducing periodic structures will enhance the control.

For the control of fluid flows, the original fluid dynamic equations – the Navier–Stokes equations – will need to be directly considered. Because of the form invariance of the simplified Navier–Stokes equation (that is, the Stokes equations), the way to manipulate electromagnetic waves based on transformation optics can thus be extended to the control of fluid flow^{38,196}. Doing so leads to the formation of a new area of hydrodynamic metamaterials¹⁹⁷. Unlike water wave manipulation effects based on the design of effective water depth or gravitational acceleration, the control of fluid flow is based on the design of viscosity. By carefully designing the effective viscosity of the fluid, the manipulation effects of steering, refracting and focusing of fluid flows can be realized¹⁹⁸. Furthermore, based on transformation hydrodynamics, functional devices for the control of fluid flows can be realized, such as hydrodynamic cloaks^{38,199}, hydrodynamic concentrators²⁰⁰ and hydrodynamic rotators²⁰¹. Because of these promising practical applications, we hope more functional hydrodynamic metamaterials based on the original fluid-dynamic equations can be realized in the future.

On the basis of the similarity between 2D Maxwell equations and water wave equations, water wave crystals and metamaterials can be designed to control water waves, which provide a simple method to obtain functional water wave devices. Although it is an approximate solution, it will facilitate understanding of water wave control in 3D. Whether the phenomena achieved under 2D approximation will appear in 3D needs to be further verified, which may lead to a new direction. We hope that there will be more artificial structures designed to control water waves based on 3D calculation methods in the future, which will further promote the practical applications.

In summary, ever since the similarity between water waves and electromagnetic waves has been discovered, the field of water wave control has been greatly developed, and many new control methods in electromagnetic waves have been introduced into water wave

control. We hope this Review will be of use to beginners interested in this field and will give inspiration to researchers in other fields. We look forward to more and more researchers joining this field and pushing the boundary of the water wave study.

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Author contributions

S.Z., X.Z. and L.H. researched data for the article. S.Z., X.Z., L.H., X.H. and H.C. contributed substantially to discussion of the content. S.Z., X.Z. and L.H. wrote the article. X.H. and H.C. conceived and supervised the whole project. All authors reviewed and/or edited the manuscript before submission.

Competing interests

The authors declare no competing interests.

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